FORECAST OBJECTS, PRINCIPLES, AND APPROACHES

Learning Outcome

- explain principles and approaches to forecasting a company's financial results and position

Approaches to forecasting vary by analyst and setting. Analysts who publish company research for external distribution tend to maintain quarterly forecasts and invest a great deal of effort to achieve forecast accuracy of revenue and earnings per share estimates. An investor with a controlling position in a private company has access to far more information, which may be used to build highly detailed models with a multi-year or even multi-decade perspective. Additionally, there are forecasting norms that vary by business model of the target company, such that models for banks appear markedly different from those for oil and gas producers.

In this module, we take the perspective of a longer-term-oriented analyst forecasting relatively straightforward public issuers—Warehouse Club Inc. and Iliso Marketplace Ltd., introduced in the prior two modules—but the principles discussed are broadly applicable. The more technical details of financial statement modeling using spreadsheets are covered in lessons in Financial Statement Analysis.

What to Forecast?

Analysts may focus on different **forecast objects** related to issuers' financial statements. Below are four common forecast objects.

1. *Drivers of financial statement lines*, which were illustrated in the earlier module on company analysis. Recall that for Warehouse Club Inc., net sales each year was analyzed using two bottom-up drivers: the average number of stores open and the average net sales per store, both of which were also analyzed using year-over-year percentage changes over the last 10 years. Forecasts of net sales (at the top of Warehouse's income statement) can be made by forecasting these drivers individually and multiplying them. Gross margin and SG&A expenses as percentages of net sales were identified as key drivers of operating expenses, which can be forecast and then multiplied by the net sales forecast to obtain the forecast cost of sales and SG&A expenses. Forecasting drivers rather than financial statement lines outright has the benefit of improved explanatory value and may improve accuracy; a financial statement line item can have multiple drivers moving in different directions, which might be difficult to forecast on an aggregated basis.
2. *Individual financial statement lines*. Rather than drivers, analysts can also directly forecast individual financial statement lines. This approach is often used for lines without clear drivers, for less-material items, and for items that the analyst does not have a perspective on. In some cases, the analyst may look for an estimate by management or simply assume the quantity will remain the same in future periods as in the past. Examples include lines such as amortization expense on the income statement (a forecast for which is often provided by management in the annual report), "other non-current assets" on the balance sheet (when small), and various lines on statements of cash flows for which minimal disclosures are provided.
3. *Summary measures* such as free cash flow, earnings per share, and total assets. The benefit of using these as forecast objects is efficiency. The disadvantage is less transparency, making it difficult to audit the forecast. This approach is most appropriate if the summary measure is stable and predictable, or if issuer disclosures are severely limited.

investment decision with respect to the company's equity or debt securities before the issuer recognizes an accrual on its financial statements. Examples of such cases include an issuer's announcing a material legal proceeding, government regulatory action, a tax dispute, or a natural disaster.

Focus on Objects That Are Regularly Disclosed

We recommend using forecast objects that either are disclosed regularly or can be directly calculated using what is disclosed regularly (e.g., for Warehouse Club: net sales, number of stores, net sales per store, national retail sales published monthly by a government agency). Information that is *not* disclosed regularly (such as the size of a market from a third-party consultancy's report) is suitable for *informing* forecasts but can be problematic for direct use because forecasts cannot be confirmed in a timely manner. While it may be intuitive to forecast sales and gross margin by individual product line for a company, if gross margin is disclosed only on a consolidated basis, it will be difficult for the analyst to verify product-line gross margin estimates. If the company's reported gross margin results differ from gross margin estimates, is the forecast off on product A, B, or C? Similarly, if a quantity is disclosed only on an annual basis, a forecast of it can be confirmed only once a year.

We also caution against overly complex models, because they require more forecasts and take more time to update, often without any improvement in accuracy. For example, the Swiss pharmaceutical company Novartis AG discloses revenues on a quarterly basis for over 35 individual drugs by US and Rest of World regions. Most of the drugs are approved for more than one type of illness. The company also has over 50 drugs in late-stage clinical development that may drive future revenue. While an analyst could forecast Novartis AG's revenues by forecasting sales for each of its individual drugs by illness and geography, it would take months to create such a model and weeks to update it every quarter, and the forecasts by illness are not verifiable over time based on the company's disclosures. Instead, the analyst should use judgment to focus on the most important drivers and use aggregations and shortcuts where the value-added of discrete forecasting is low. For example, it would not be a good use of an analyst's time to forecast sales for drugs with small addressable markets or those that have reached maturity and will exhibit slow growth until they lose patent protection.

Forecast Approaches

For any object, there are four general forecast approaches, as shown in Exhibit 1. In practice, they are often combined.

Exhibit 1:

Forecast Approaches for Financial Statement Objects

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Historical Results: Assume Past is Precedent

This approach uses past observed or calculated values (such as a historical median) as a forecast. This is the easiest approach and is, perhaps, the default. Because the past occurred at least once before, it seems plausible to expect that it will again. However, past results were produced in past conditions, which may differ from current and future conditions.

Using Historical Results as Forecasts

An analyst is forecasting the market size for energy drinks, a type of non-alcoholic ready-to-drink beverage. The analyst uses two drivers: a forecast of the market for non-alcoholic ready-to-drink beverages and energy drinks' share of that market. Sales of non-alcoholic ready-to-drink beverages have grown, on average, by 3% per year for the last 10 years. Since the analyst does not expect any material changes in conditions, she forecasts that the market will continue to grow by 3% per year.

The analyst believes that energy drinks bear some similarities to sports drinks, another type of non-alcoholic ready-to-drink beverage, which has achieved a market share of 20%. The analyst thus forecasts that energy drinks will also reach a market of 20% by 2035. These two forecasts are multiplied to arrive at a forecast market size for energy drinks.

This approach may be appropriate for companies operating in industries where the analyst does not expect the industry structure (e.g., Porter's Five Forces, PESTLE influences) to change, as well as for companies that have a low sensitivity to changes in the business cycle. This approach is also commonly used for forecast objects that are not material or that the analyst does not hold an opinion on.

Historical results are a less appropriate forecast approach for companies in cyclical industries, because a future period is likely to be at a different point in the business cycle than the current or past period, and so results will differ. An "over the cycle" average or median may be appropriate for a multi-year forecast for a cyclical company, but less appropriate for a specific year as such an average or median disguises the year-to-year volatility. This forecast approach is also less appropriate for companies that are changing their competitive strategy or restructuring in some way, such as making a large acquisition or divestiture that renders historical results non-comparable.

period of time, as a "base rate" for forecasting that an object will converge to over some time frame. The base rate could also be a macroeconomic variable such as GDP growth. This approach is not a rote historical exercise, as analyst discretion is required in selecting the object, the sample to calculate the base rate, and a time frame for convergence to the base rate.

Historical Base Rates and Convergence

Tesla is a relatively new entrant in the automotive industry, with an innovative suite of electric vehicles. An analyst may forecast Tesla's income statement by forecasting that its sales growth, gross and operating margins, capital intensity, and capital structure will converge to historical automotive industry averages (using a peer group of publicly traded global automakers) over the next 10 years.

This forecast may be contested by other analysts who believe that the shift from internal combustion to electric vehicles, Tesla's innovative battery technology, and its direct-to-consumer sales approach make historical auto industry averages irrelevant for Tesla.

Historical base rates and convergence may be appropriate for forecasting companies in well-established industries with many publicly traded peers, such as banks, airlines, restaurants, automakers, and retailers, especially when the analyst's industry analysis does not call for material changes in industry structure or external industry influences. This approach is also logical for smaller companies that are "maturing into" a financial profile similar to that of larger peers with scale. For example, a biotechnology company that recently launched its first drug likely spends more than half its revenues on R&D, a ratio that is likely to decline toward the industry norm as the company scales.

This approach is less appropriate for companies in changing or new industries for which calculating a base rate is difficult or less relevant, for companies in highly cyclical industries for which a longer-term base rate and smooth convergence to it would obscure year-to-year volatility, and for companies that are industry leaders by a wide margin and account for a substantial proportion of any industry base rate. For example, "digital advertising industry" averages for sales growth and operating margins would be dominated by Alphabet and Meta, such that using them to forecast Alphabet's and Meta's financial results would simply be using their own historical results as the forecast, which may ignore any benefit from economies of scale (as the companies were smaller in the past).

Management guidance

Management of public companies may publicly provide targets for earnings, revenues, and other measures (e.g., capital expenditures) for the next quarter, year, or longer term, known as **management guidance** (or simply as guidance). Guidance can be detailed or rather directional and is often updated throughout the year. Initial guidance for next fiscal year might be provided during the fourth-quarter earnings call and updated for completed quarters, and new information provided at the first-, second-, and third-quarter earnings calls. **Beyer et al. (2010)** found that guidance accounts for a majority of the quarterly financial reporting information used by investors. Its importance stems from its forward-looking rather than backward-looking nature, and because management has an abundance of industry and company information that investors do not.

Guidance is often provided as a range (e.g., "sales growth of 2%–4%") and embeds many sub-forecasts and assumptions by management, including macroeconomic growth, cost inflation, market share changes, pricing actions, and changes in exchange rates. If management's estimates for these change, so will guidance, and it is not uncommon for companies to suspend guidance altogether in periods of high uncertainty, such as during

While the midpoint of a guidance range might be intuitively interpreted as management's "true" expectations, **Ciconte et al. (2013)** found that the upper bound often better represents management's expectations, which are "padded" by associating it with a pessimistic lower bound to create a hurdle that is more easily cleared, for which management can be rewarded.

Short- and Long-Term Management Guidance

Novo Nordisk A/S is a large biopharmaceutical company with shares and American Depositary Receipts (ADRs) listed on the NASDAQ Copenhagen and NYSE, respectively. Its management provides annual guidance that it updates each quarter, as well as less detailed but longer-term "strategic aspirations" during its annual Capital Markets Day presentations. Guidance includes management's point and range estimates for the following (for each fiscal year):

- Sales growth
- Operating profit growth
- Net interest income/expense
- Effective tax rate
- Capital expenditures
- Depreciation and amortization
- Free cash flow

Using guidance for forecasts is appropriate when it is provided and when management has demonstrated a track record of reliable estimates (analysts should analyze past guidance versus actuals). We caution against the use of guidance for companies that are highly sensitive to the business cycle, as management does *not* have an informational advantage over investors at forecasting macroeconomic variables like GDP or the prices of commodities. **Hutton et al. (2012)** found that investors have an informational advantage at the macroeconomic level, while management has an advantage at the firm level and tends to make more accurate forecasts for objects under greater management control, such as operating expenses and capital expenditures.

Analyst's discretionary forecast

We group all other forecast approaches as analysts' discretionary forecasts, which include those based on surveys, quantitative models, probability distributions, analogies to historical precedents that differ from comparable companies or industry averages, and other unobservable inputs. This approach is most common for companies in cyclical industries, companies that have no or few comparables, those that do not provide management guidance, and/or those undergoing a fundamental change like a shift in the competitive or regulatory environment. For example, while shifts in energy production have happened before (e.g., coal to natural gas), climate change and the potential shift to renewable energy and adoption of technologies like electric vehicles are unprecedented, with no historical observations to use as forecasts. Instead, an energy sector analyst must make discretionary forecasts based on many sources, such as government emission reduction commitments, proposed legislation, capital expenditure constraints, and so on.

Discretionary Forecasts

An analyst is building a financial statement model for a medical technology company developing a new device to treat a type of stroke for which no other effective treatments exist. The analyst uses information from the device's clinical trials and epidemiology data from various medical journals to estimate the total addressable market of patients that are treatable with the device. The analyst also conducts a survey of specialist physicians to gauge their interest in using the device and to

cardiovascular diseases to estimate market share for this device for the next 10 years. The analyst develops several market share scenarios based on the results of the clinical trials (i.e., the safer and more effective the device, the higher sales growth will be). While the analyst uses comparable, historical results as inputs, significant judgment is also required because this situation is unique.

Selecting a Forecast Horizon

The choice of the forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicality of the industry, company-specific factors, and the analyst's employer's preferences. Most professionally managed investment strategies describe the investment time frame and average holding period in the investment objectives of the strategy. For example, long-term-oriented fund managers may focus their forecasting primarily on a time period of three to five years, while shorter-term-oriented managers may focus more on the next one or two quarters.

The cyclicality of the industry could also influence the analyst's choice of time frame, because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability. Similar to cyclicality, various company-specific factors, including recent acquisition or restructuring activity, can influence the selection of the forecast period to allow enough time for the realization of the expected benefits from such activity to be reflected in the financial statements. In other cases, there might be no individual analyst choice in the sense that the analyst's employer has specified fixed parameters.

Question Set

1. A company discloses financial results by geographic area, and the economic exposure of each geographic area is similar. Explain why an analyst should or should not develop forecasts by geographic area and aggregate the results.

Hide Solution

Solution:

An analyst should *not* develop forecasts by geographic area, because each area has similar economic exposure. Developing forecasts for each geographic area would require more time to develop and update, without a significant improvement in accuracy over an aggregate forecast.

2. Which approach is *most* appropriate for an analyst to use to forecast revenue for an industry leader in a mature, non-cyclical industry?

A. Historical results
B. Analyst's discretionary forecast
C. Historical base rates and convergence

Hide Solution

Solution:

A is correct. Historical results is appropriate for companies operating in mature industries where the analyst does not expect the industry structure to change and for companies with low sensitivity to the business cycle. Analyst's discretionary forecast is most common for companies that have no or few comparables, do not provide management guidance, and/or are undergoing a fundamental change. Historical base rates and convergence is not appropriate, because the company is an industry leader and would account for a substantial proportion of any industry base rate.

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A. The availability of guidance increases during periods of high uncertainty.
B. The midpoint of guidance best represents management's "true" expectations.
C. Understanding the assumptions behind the guidance is critical to assessing the guidance.

Hide Solution

Solution:

C is correct. A key focus of an analyst is to understand management's assumptions embedded in guidance and to scrutinize their plausibility.

It is not uncommon for companies to suspend guidance altogether in periods of high uncertainty. While the midpoint of a guidance range might be intuitively interpreted as management's "true" expectations, management may provide a range specifically intended to create a hurdle that is more easily cleared, for which management can be rewarded.

4. Which of the following is an example of an ad hoc forecast object?

A. Amortization expense
B. Average sales per store
C. Pending legal proceedings

Hide Solution

Solution:

C is correct. Ad hoc objects are those that may not yet be reflected in historical financial statements. Amortization expense is an individual financial statement line. Average sales per store is a bottom-up driver of a financial statement line (revenue).

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